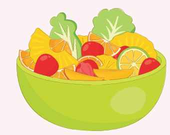
Flowcharts

05

A flowchart helps us to design and document a process. Before we perform a task, a flowchart helps us to plan it out. It shows the steps of the process and the flow of the task. It helps us to visualise the process and check for any flaws.

For example: To make fruit salad, we have to follow step by step instructions:



Step 1

Gather the ingredients required:
2 mangoes, 1/2 cup of lemon juice, 1/2 cup of fresh pineapple, 3 fresh strawberries, 1 cup of orange juice, 1 bowl.

Step 2

Check if the fruits are clean. If not, wash them.

Step 3

Cut the fruits into small cubes.

Step 4

Mix lemon juice with orange juice and pour over the cut fruits.

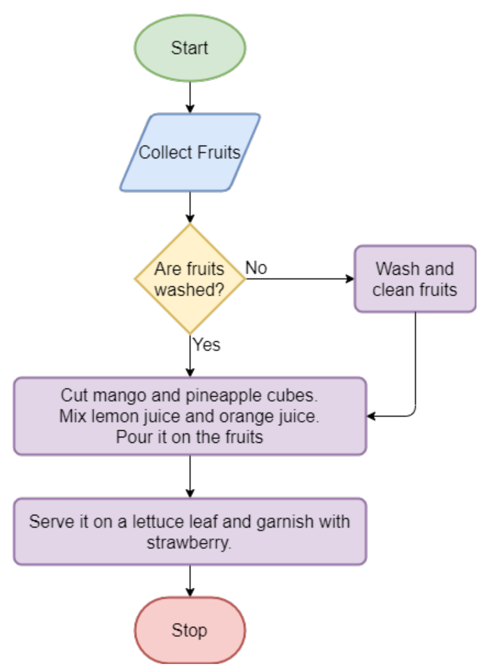
Step 5

Serve on a bowl and garnish with strawberries.

The process of preparing a fruit salad can be shown as a flowchart. This will make it organised and easier to understand.

Flowchart

- A flowchart is a diagram that represents a process or a step-by-step approach to perform a task.
- A flowchart typically includes a series of activities, inputs, outputs, and roles (the people that perform the activities) in a sequence.



Bytes

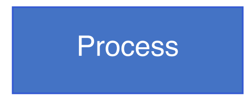
A process is a series of actions or steps taken in order to complete a particular task.

5. Flowcharts

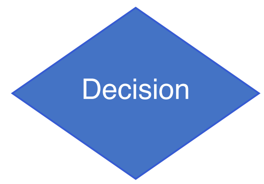
A flowchart consists of 4 main elements which are connected with arrows to show the direction of flow:

Terminator: The terminator symbol represents the start or end point of the flowchart.

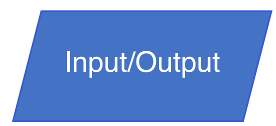
Process: A box indicates a specific task.



Decision: A diamond represents a decision point. The lines that come out from the diamond indicate different possibilities, which lead to different sub-processes.



Data: A parallelogram represents input and output. This box receives data and displays processed data.



Flow: Arrows represent the flow of the sequence and the direction of a process.

Uses of a flowchart

- To simplify complex processes.
- To make documenting a process quick and clear.
- To develop an understanding of how a task is done.
- To study a process for improvement.
- For better communication when one or more people are involved in the same task.

Consider the following points before creating a flowchart

- The parameters of the process: Where or when does the process start? Where or when does it end?
- Discuss and decide the activities that take place as part of the process.
- Arrange the activities in a proper sequence.
- When the sequence is correct, draw arrows to show the flow of the process.
- Review the flowchart to check if the process is accurate.

Exercise

Fill in the blanks

1. A ____________________ is a diagram that represents a process or a step-by-step approach to complete a task.
2. ____________________ is a series of actions or steps taken in order to achieve a particular task.

State whether True or False

3. A diamond shape flowchart element represents a process point.

▲ ________________________

4. Flowchart helps makes documenting a process quick and clear.

▲ ________________________

Let's learn more about flowcharts through this simple activity:
Pre-Activity: Draw a flowchart of your daily morning routine in your notebook.

Sample answer:

Brush Teeth → Get Dressed → Eat Breakfast → Get backpack ready → Go to School

We can draw a flowchart on paper with pencil or pen. We can also create it on a computer. The applications that can be used to create flowcharts are One Note, Draw.io, Word, PowerPoint and Google Draw.

Now, let's try to create another flowchart on the computer.

Activity

- Create a flowchart to show the process to identify objects that float or sink in water. Use the Draw.io application to create it.

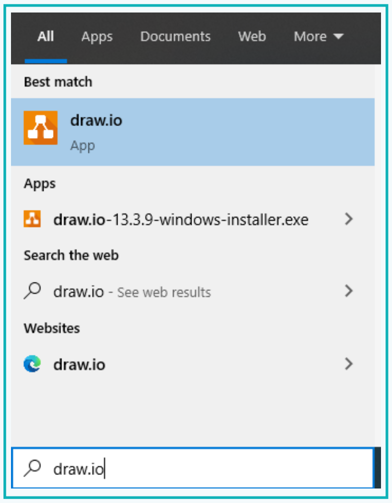
To identify which of the given objects will float or sink, we will drop them one by one into a bucket of water. Then we will write our observations in the table below. We will create a flowchart of the process in the **Draw.io** app.

Objects	Sink or Float
Wooden Stick	
Rock	
Coin	
Balloon	
Sponge	

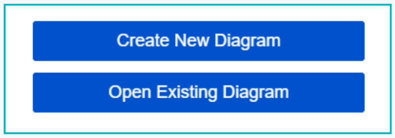
Sinking and Floating

Activity

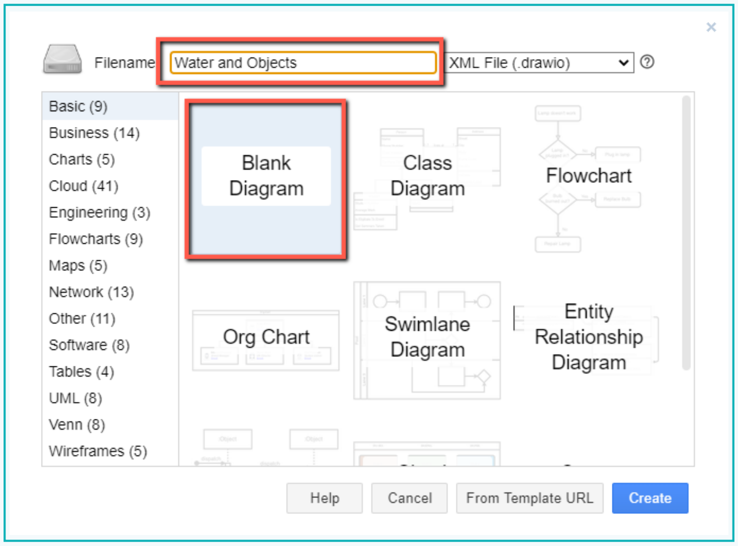
1. Search for the 'Draw.io' app from the taskbar and launch it.



2. Search for the 'Draw.io' app from the taskbar and launch it.



3. Add a file name. We can call this project "Water and Objects".
   a. Select 'Blank Diagram'.
   b. Click on 'Create'.



To start the flowchart, we will use the 'Terminator' symbol.

Activity

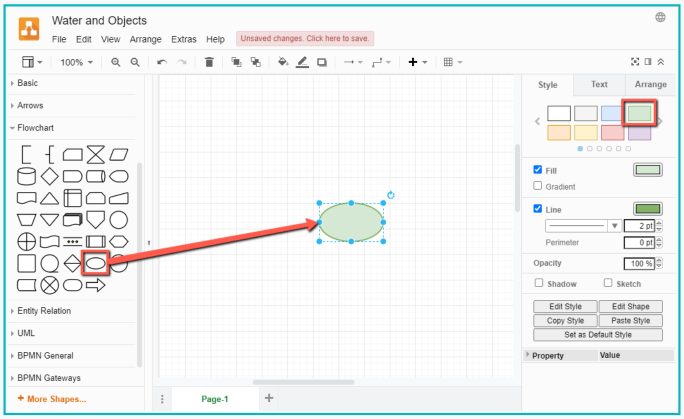
4 Create the ‘Terminator’ symbol to indicate the start of the flow.

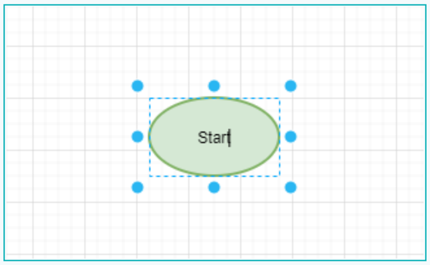
a. From the ‘Flowchart’ palate, place a ‘Start’ element.

Colour coding the elements helps to easily identify those elements.

b. Select a green colour from the ‘Style’ palate.

c. Select the element and type ‘Start’ inside it.



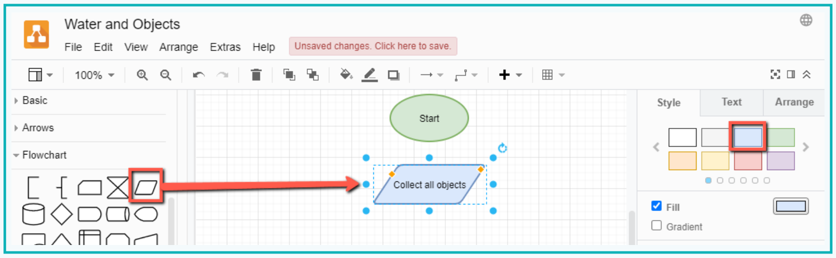


To identify which of the objects will float or sink in water, we need to collect all of them.

5 Show the objects or data involved in the process with the ‘Data’ element.

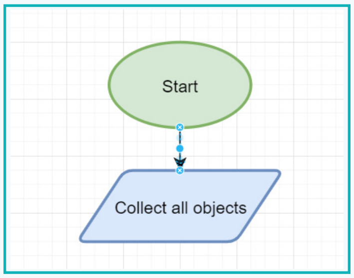
a. Place the ‘Data’ element below Start. Name it ‘Collect all object’.

b. Colour code it blue.



6 Show the process flow.

a. Use the Flow element ‘Arrow’ to connect elements with each other.



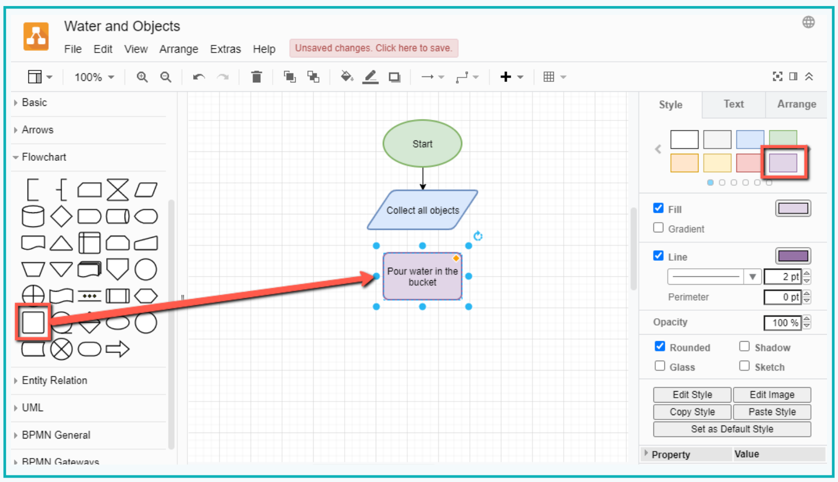
Activity

After we have collected all the objects, we need to fill a bucket with tap water.

7 Show the process to fill a bucket with water.

a. Place a ‘Process’ element.

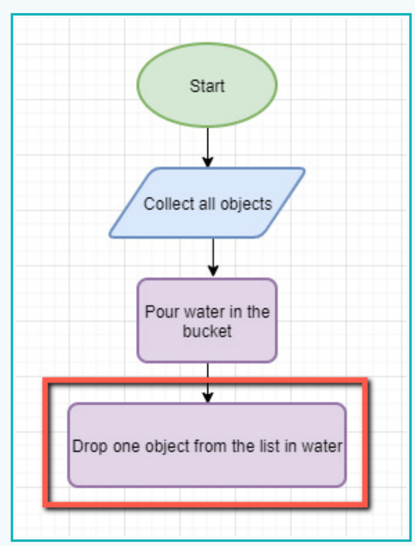
b. Type ‘Pour water in the bucket’ as the next step.



After we have filled the bucket with water, we drop the objects in one-by-one to see if they float or sink.

8 Show the next step with another ‘Process’ element.

a. Type text ‘Drop one object from the list in water’.

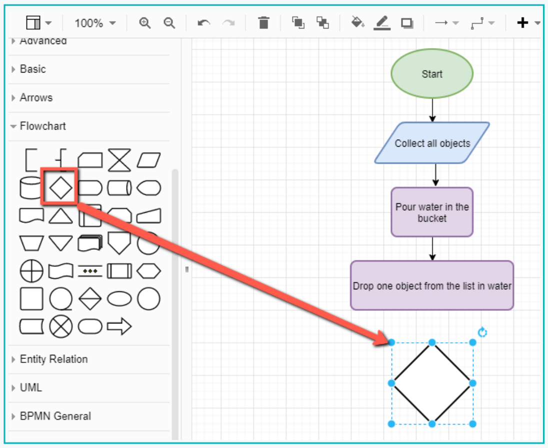


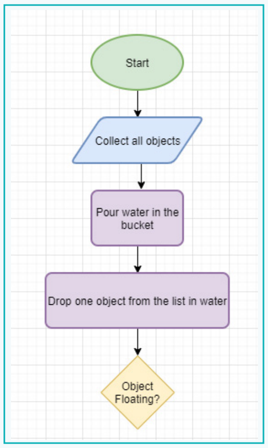
After we drop the object, we check if the object floats or sinks and make our decision.

Activity

9 Decide whether to write float or sink.

a. Place the 'Decision' element and type 'Object Floating?'. Colour code it yellow.





'Decision' element will have two cases.

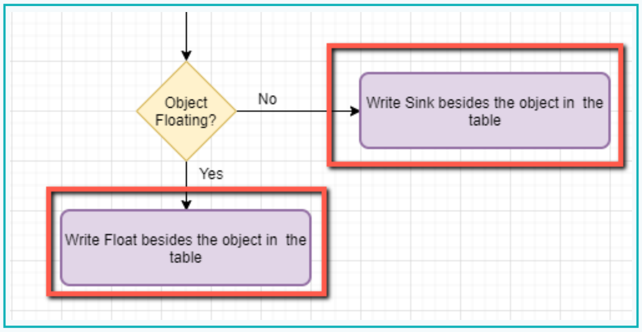
The Decision element will check if the object floats or not. Based on this, the next step will be decided.

- If the object floats, we will write 'Float' next to the object name in the table.
- If the object sinks, we will write 'Sink' next to the object name.

10 Make a decision.

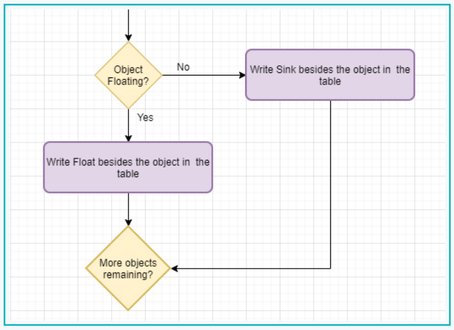
a. If the object floats, we will write 'Yes' on the flow arrow. Add a process element and type 'Write Float in the column beside the object in the table'.

b. If the object sinks, we will write 'No' on the flow arrow. Add a process element and type 'Write Sink in the column beside the object name in the table'.



Activity

Now we check whether any other objects are left to drop into water.

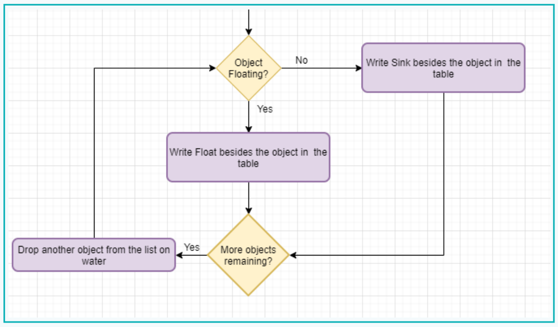


11 Add another decision element to check if any objects are left.

a. Type ‘More Objects remaining?’.

b. Colour code it yellow.

If more objects remain, we perform the whole process again.



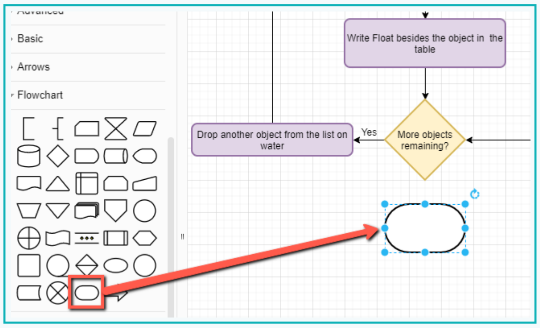
12 If more objects left to be tested:

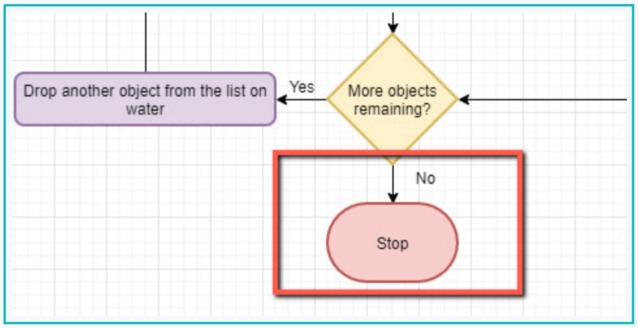
a. Type ‘Yes’ on the flow arrow. Add a process element and type ‘Drop another object from the list in water’. Connect it to ‘object floating’ to check if it floats or sinks.

If no more objects remain, we can stop the process.

13 Stop the process.

a. Place the ‘Terminator’ element and type ‘Stop’ inside it.

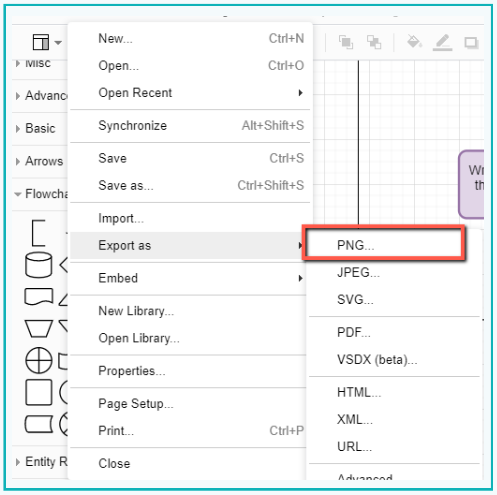




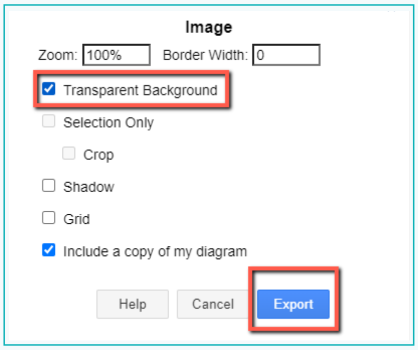
We will save the file as an image with a transparent background.

Activity

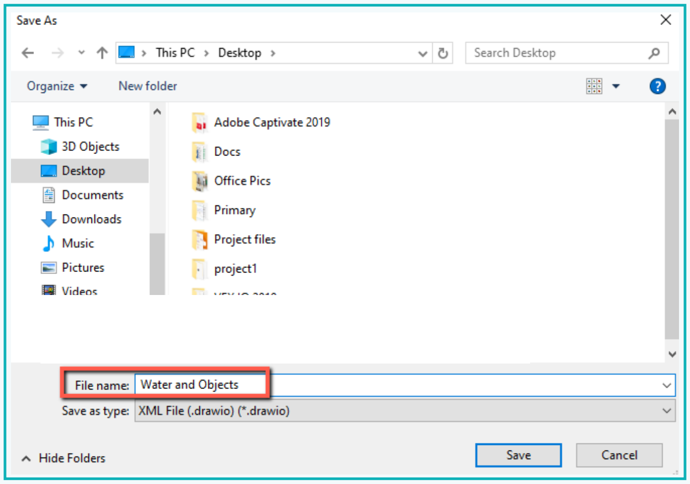
14 Export the image as .png. This will let us use it in any type of document.



15 Select 'Transparent Background' to make the background transparent.

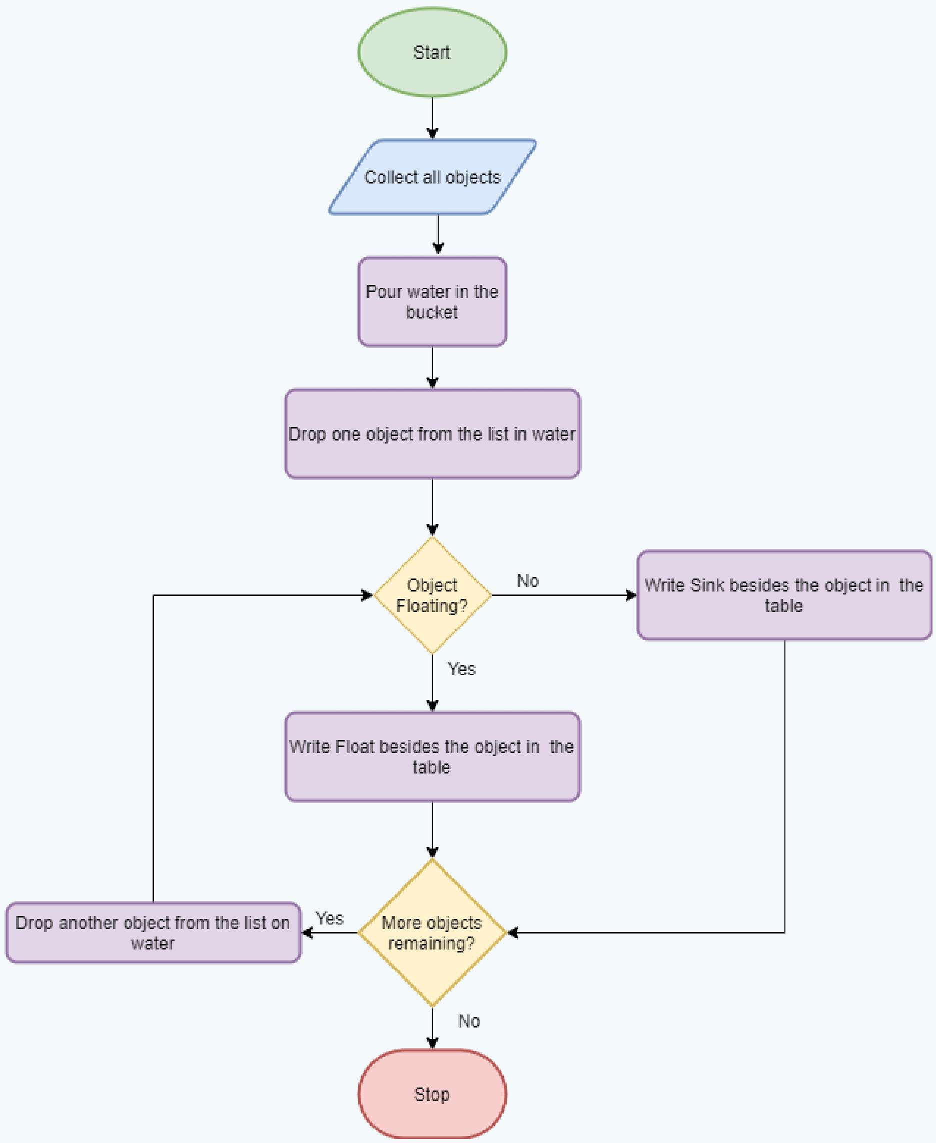


16 Save the file as shown.



Activity

Final Output:



This is what our flowchart will look like. It shows the process to identify objects that float or sink in water.

A flowchart helps us to design and document a process. It helps us to visualise and understand a process, and find the flaws in it. It is created with arrows, symbols, and few words. It helps to analyse the process and check all possible inputs and outputs.

Tech Flash

- **Flowchart** : A flowchart is a diagram that represents a process or a step-by-step approach to complete a task.
- **Terminator** : The terminator symbol represents the start or end point of a flowchart.
- **Process** : The process box indicates a specific task.
- **Decision** : A diamond shape represents a decision point. Lines coming out from the diamond indicates different possible outcomes, which lead to different sub-processes.
- **Data** : Data represents input and output. This box receives data and displays processed data.
- **Flow** : An arrow represents the flow of the sequence and direction of a process.